AI COMMUNICATION SYSTEMS STUDY KIT

PROMPT LIBRARY

Ready-to-use prompts for learning, simulating, testing, and presenting AM/FM communication concepts.

Student: _____

Department: Electronics and Communication Engineering (ECE)

College: _____

How to use: Copy one prompt into an AI assistant or use it as a task description. Always verify technical outputs and include only work you actually completed.

1. Beginner learning prompts

Prompt 1. Explain amplitude modulation (AM) in simple beginner-friendly English, with a real-life example, equation, and description of its waveform.

Prompt 2. Explain frequency modulation (FM) in simple beginner-friendly English. Explain what changes and what stays constant.

Prompt 3. Compare AM and FM in a table using carrier property, bandwidth, noise performance, complexity, and applications.

Prompt 4. Explain message signal, carrier signal, modulation, modulation index, frequency, amplitude, and bandwidth with short definitions.

2. Python simulation prompts

Prompt 1. Write beginner-friendly Python code using NumPy and Matplotlib to generate a message signal and a carrier signal. Label the axes and add a grid.

Prompt 2. Write Python code to simulate amplitude modulation (AM). Use clear parameter names, plot the message and AM waveform, and explain each line.

Prompt 3. Write Python code to simulate frequency modulation (FM). Plot the message and FM waveform, and explain why the waveform spacing changes.

Prompt 4. Combine message, carrier, AM, and FM plots into one script. Save each graph as a PNG file in a folder named communication_graphs.

Prompt 5. Review this AM/FM simulation code for syntax errors, incorrect equations, and misleading graph labels. Explain each correction for a beginner.

3. Automation prompts

Prompt 1. Design a simple workflow that runs an AM/FM simulation, generates four plots, and saves the plots automatically. List prerequisites and steps.

Prompt 2. Explain the difference between automation and artificial intelligence using my AM/FM study-kit project as the example. Do not claim AI features that are not implemented.

Prompt 3. Create a checklist to test that the generated message, carrier, AM, and FM graphs are saved correctly and show expected behavior.

4. Study-kit and quiz prompts

Prompt 1. Create a glossary of 15 important communication-system terms with short, beginner-friendly definitions.

Prompt 2. Create 10 multiple-choice questions about AM and FM, with four options each, an answer key, and one-sentence explanations.

Prompt 3. Prepare 8 viva questions and short answers for an ECE student presenting an AM/FM simulation project.

Prompt 4. Make a one-page revision sheet covering AM equation, FM equation, key differences, and applications.

5. Report and presentation prompts

Prompt 1. Draft a project report for an AI Communication Systems Study Kit that includes aim, objectives, tools, methodology, outputs, limitations, conclusion, and future improvements. Mark any details I must fill in myself.

Prompt 2. Create a 10-slide presentation outline for an AM/FM simulation study kit. Keep each slide concise and suggest where actual generated graphs should be inserted.

Prompt 3. Write a README.md for a GitHub repository containing Python AM/FM simulation code, generated graph images, a report, and a presentation. Use placeholders for unknown links and student details.

6. Debugging and verification prompts

Prompt 1. Explain this Python error in beginner-friendly language and give the smallest fix: [paste error message and relevant code].

Prompt 2. Check whether the AM waveform envelope follows the message signal and whether the FM waveform changes instantaneous frequency while keeping ideal amplitude constant. Tell me what evidence to inspect.

Prompt 3. Give me a final submission checklist for code, graphs, report PDF, presentation PDF, README, and public repository link.

Important notes

• A prompt is an instruction, not proof that the task has been completed.
• Verify equations, code, plots, and references before submitting.
• If your project is described as an AI study kit, clearly state which parts actually use AI and which parts are ordinary Python simulation or automation.
• Replace all placeholders with your real details; never invent results or repository links.